

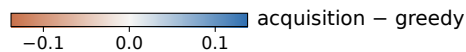
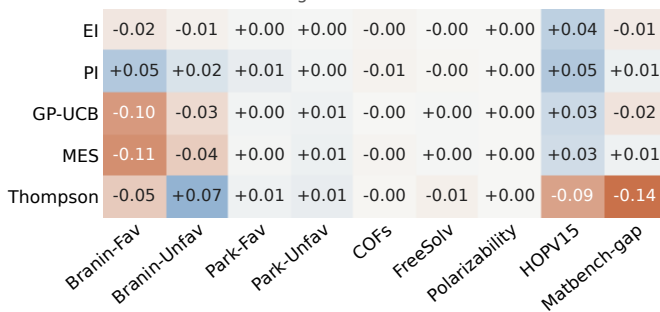
a Share of TL surrogates attaining more than greedy

5 surrogates · 20 seeds each



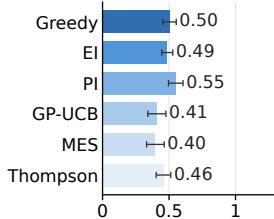
b Mean attainment difference vs greedy

mean over the 5 surrogates



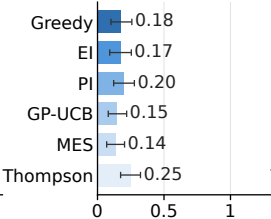
c Brannin-Fav

B = 50



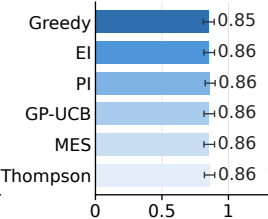
d Brannin-Unfav

B = 50



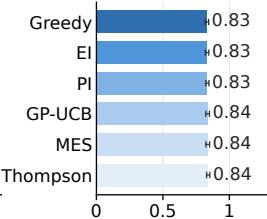
e Park-Fav

B = 50



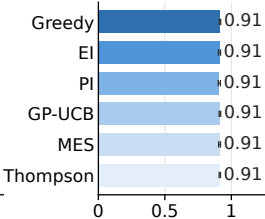
f Park-Unfav

B = 50



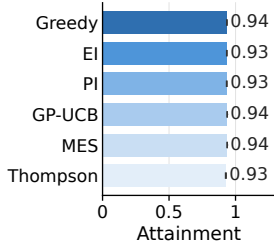
g COFs

B = 30



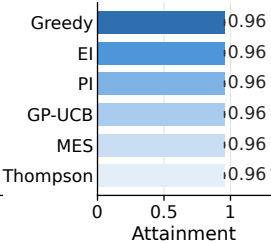
h FreeSolv

B = 30



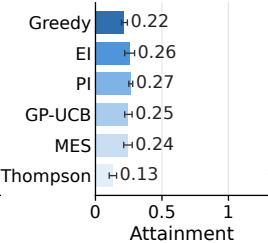
i Polarizability

B = 30



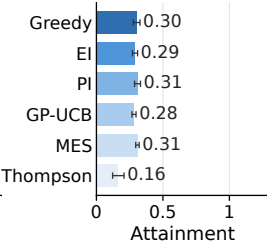
j HOPV15

B = 30



k Matbench-gap

B = 20



c-k: mean $\pm$ s.e. over the five TL surrogates of each surrogate's 20-seed mean attainment (seeds 42-61).

a, b: paired per surrogate (acquisition - greedy), then the share of positive differences (a) or the mean difference (b).

TL: TL-base = base transfer-learning surrogate · TL-PtJ = pretrain-then-joint · TL-E2E = end-to-end joint

TL-SPS = soft parameter sharing · TL-MMD = domain adaptation (MMD)

GP: GP-base = autoregressive multi-fidelity GP (MFGP) · GP-SV = sparse variational MFGP · GP-DKL = deep-kernel GP